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Plugging Characteristics of Temporary Plugging Materials at Multi-Perforation Holes in Multiple Clusters Along the Wellbore: A Novel Experimental Study

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Abstract

Conducting perforation temporary plugging experiments and analyzing the plugging characteristics of temporary plugging materials at the perforations are of great significance for the temporary plugging and diversion fracturing in shale gas reservoirs. However, the current experimental apparatus merely contains a short horizontal wellbore with a limited number of perforation holes, without accounting for the erosive effect of proppant on the holes. This shortage makes it difficult to the actual plugging characteristics of temporary plugging materials at multiple perforation holes in multiple clusters along the horizontal wellbore. In this study, a new experimental system for temporary plugging of perforation holes is established, involving a 10-m horizontal wellbore with 6-meter cluster spacing two perforation clusters, and 6 perforation holes per cluster. The average injection rate can reach 66 L/min per perforation, replicating the high pumping rate in the real field. The shale of perforation holes are adjustable and could be either circular or teardrop shapes with equivalent diameters ranging from 12 to 16 mm. This experimental system is used to study the plugging characteristics of various temporary plugging materials at different perforation locations along the wellbore under different dosages and application strategies. The results show that: (1) perforation holes near the well toe and at the wellbore bottom are most likely to be plugged, while those near the well heel and at the wellbore top have the minimal possibility to be plugged. (2) Under the same dosage of temporary plugging balls, adding them in two batches consecutively results in higher plugging efficiency compared to a single-batch addition. (3) As the dosage of temporary plugging balls increases, the number of plugged perforation holes initially rises and then stabilizes. The optimal number of temporary plugging balls is 1.2 to 1.5 times the number of open perforations. (4) Teardrop-shaped perforation holes still allow fluid inflow after being plugged by temporary plugging balls, requiring the use of a temporary plugging agent for complete sealing. However, temporary plugging knots can fully seal teardrop-shaped perforation holes. The experimental system developed in this study closely replicates real-field conditions. The insights obtained from this study have been widely applied in the Weiyuan Shale Gas Field, Sichuan Basin, China. Taking W204H49-3 well

as an example, the temporary plugging knots were applied at a dosage of 1.2–1.5 times the number of open perforations in two fractured stages with a 30–50% increase in the perforation hole opening rate.

Introduction

Multistage multi-cluster fracturing in horizontal wells is pivotal for economical shale gas development. However, reservoir heterogeneity leads to incomplete cluster activation and inadequate lateral coverage during stimulation (Xu et al. 2018; Zhao et al. 2021). Temporary plugging and diversion fracturing, which seals opened perforations to force fluid into unstimulated clusters, provides a critical solution (Ataur R et al. 2017; William et al. 2018; Guo et al. 2023). While extensive research exists on initiation and propagation of new fractures post-plugging (Wang et al. 2018; Wang, D et al. 2020; Tang et al. 2020; Chang et al. 2024), studies on plugging characteristics of temporary plugging materials at multi-perforation holes in multiple clusters along the wellbore remain limited.

The CFD-DEM (Computational Fluid Dynamics-Discrete Element Method) numerical simulation serves as the predominant methodology for current research on sealing mechanisms of temporary plugging balls at perforations. Utilizing this approach, scholars have systematically evaluated the impact of multiple critical parameters—including ball position, density, dosage, carrier fluid injection rate, viscosity, and perforation cluster fluid intake efficiency—on perforation plugging efficiency. Zeng et al. (2024) demonstrated that closer proximity of balls to perforations increases their likelihood of entering diversion flow paths, thereby enhancing plugging efficiency. Qu et al. (2024) revealed that toe-end perforation clusters are more readily plugged due to ball inertia and turbulent effects, subsequently developing a machine learning-based predictive model for ball sealing. Hu et al. (2024) established three key findings: (1) Near-neutral buoyancy (ball density ≈ fracturing fluid density) promotes plugging; (2) A 2:1 ball-to-perforation ratio optimizes coverage; (3) Elevated viscosity increases toe-end but decreases heel-end cluster plugging probability. Lu et al. (2025) identified a positive correlation between perforation flow coefficients and plugging efficiency, establishing a diagnostic chart for ball sealing. Wang et al. (2025) quantified an optimal carrier fluid injection rate of 10-12 m^3/min maximizing plugging efficiency (showing initial increase then decrease with rate escalation), while determining a critical ball-to-perforation diameter ratio of 1.9 for peak performance. Li et al. (2025) reported maximum plugging efficiency at carrier fluid viscosities of 10-25 mPa·s. Additionally, Li et al. (2005) observed inverse proportionality between cluster density and plugging efficiency. Liu et al. (2025) extended CFD-DEM modeling to vertical wells, demonstrating that inter-ball collisions alter positioning trajectories and consequently modify sealing characteristics at perforations.

Experimental studies on perforation plugging mechanisms remain limited. Nozaki et al. (2013) integrated laboratory and field data to analyze the effects of wellbore orientation, diverting ball density, and carrier fluid injection rate on plugging efficiency. Their research identified: (1) Similar sealing behaviors in deviated and horizontal wells; (2) Buoyant balls preferentially plugging top-side perforations in horizontal wellbores, while nonbuoyant balls tend to seal bottom-side perforations; (3) Enhanced plugging efficiency with near-neutral buoyancy. Yuan et al. (2021) demonstrated through a 5.375 m horizontal wellbore apparatus (14 perforations at 30°/90°/150° phasing) that gravity high-side perforations exhibit the lowest plugging efficiency due to gravitational segregation of balls, while elevating carrier fluid viscosity, implementing mixed-density balls, and adopting downward-oriented perforating significantly enhance circumferential plugging uniformity

Current research on plugging behavior predominantly utilizes numerical simulations for temporary plugging balls, with minimal experimental focus on temporary plugging knots or temporary plugging particulate. Existing experimental setups are limited by short wellbore lengths, low perforation density, and unaddressed proppant erosion effects during fracturing. To bridge these gaps, this study developed a 10-m horizontal wellbore apparatus featuring 6-m cluster spacing, two clusters (6 perforations/cluster), and circular/teardrop-shaped perforations (12-16 mm equivalent diameter). This platform enables systematic characterization of multi-material plugging behavior across multi-cluster perforations under varied dosages and placement methods.

Experiment Design

Experimental Materials

Experimental materials comprised field-grade temporary plugging materials (Fig. 1) commonly deployed in Sichuan Basin shale gas fracturing operations: 15 mm diameter temporary plugging balls (1080 kg/m^3), temporary plugging knots with 15 mm equivalent diameter (1050 kg/m^3), and temporary plugging particulate (1–3 mm particle size, 1120 kg/m^3), with a 3 mPa·s viscosity carrier fluid enabling effective horizontal wellbore transport of all materials.

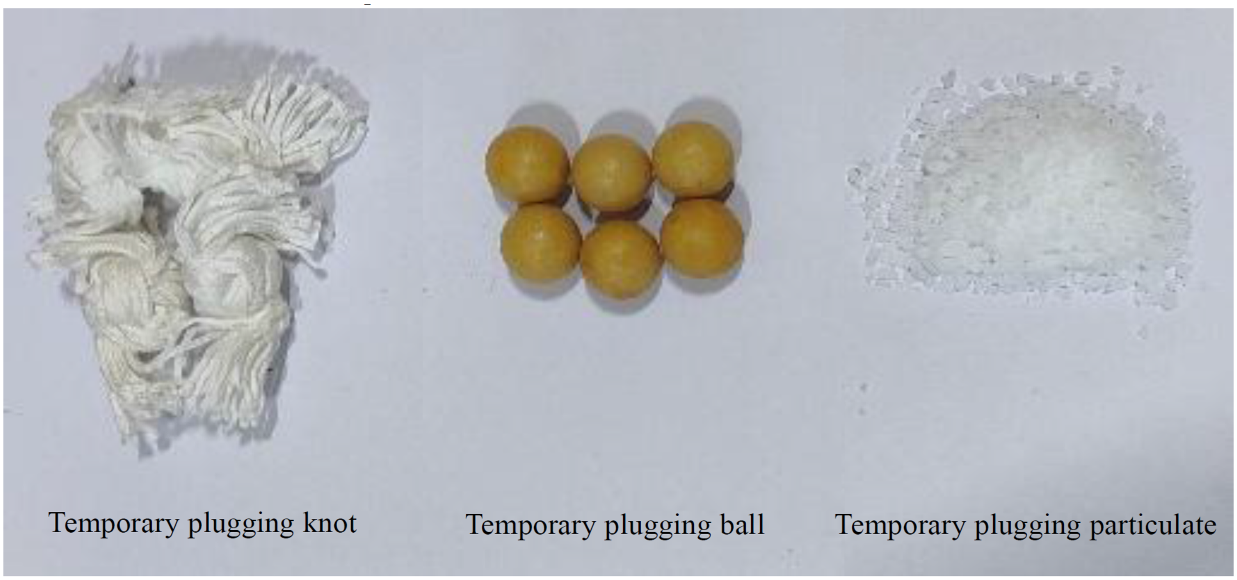

Figure 1—Temporary plugging materials

Experimental Device

This study employed a proprietary horizontal wellbore apparatus with nonuniformly eroded perforations (developed by CNPC Chuanqing Drilling & Engineering Company, Fig. 2), comprising: (1) a pumping system (500 L/min max flow rate, 20 MPa pressure capacity), (2) a control/data acquisition system enabling real-time monitoring of individual perforation injection rate and wellbore inlet pressure with automated start/stop functions, and (3) a horizontal wellbore system featuring a 10 m horizontal wellbore with two perforation clusters (Cluster 1: toe-side, Cluster 2: heel-side, 6 m cluster spacing), each containing configurable circular/teardrop-shaped perforations (12-16 mm equivalent diameter, 60° phasing, 6 max/cluster), integrated with a temporary plugging materials injection system.

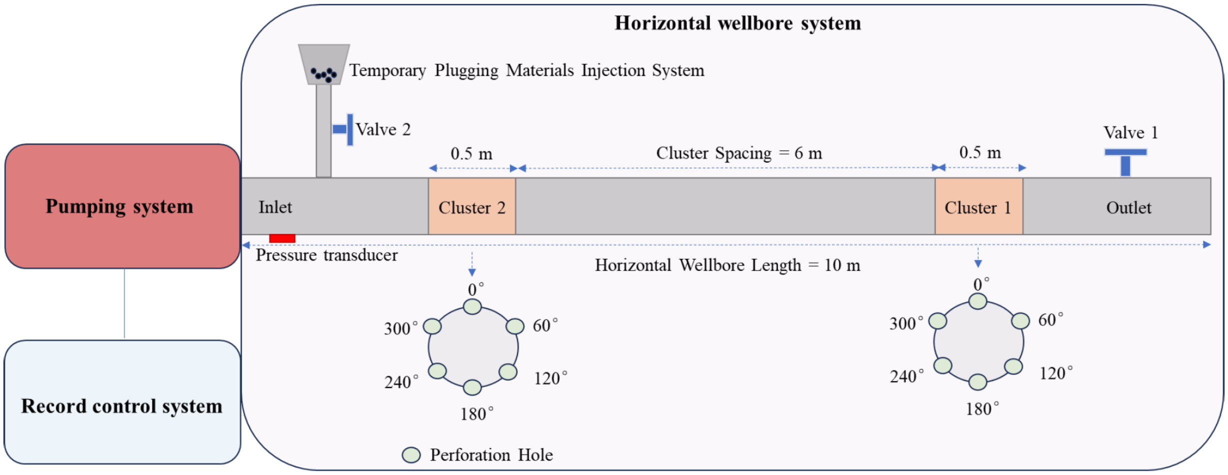

Figure 2—Experimental Device Structure Diagram

Field diagnostics in Sichuan Basin shale fracturing treatments measured pre-diversion perforation opening rates at 40-60% (He et al. 2024), accordingly, this experiment configured Clusters 1 and 2 by deactivating 0°, 240°, and 300° phased perforations while activating 60°, 120°, and 180° phased perforations, yielding six functional perforations. Downhole imaging confirmed proppant-induced erosion altered perforation geometries, generating both circular and noncircular profiles with severity escalating toward the wellbore bottom (Cramer et al. 2019; Roberts et al. 2020; Zang et al 2022). To replicate these field observations: 60° phased perforations were configured as circular (10.0 mm diameter), 120° as teardrop-shaped (12.0 mm equivalent diameter), and 180° as teardrop-shaped (14.0 mm equivalent diameter) - with Fig. 3 documenting this erosion-simulated configuration.

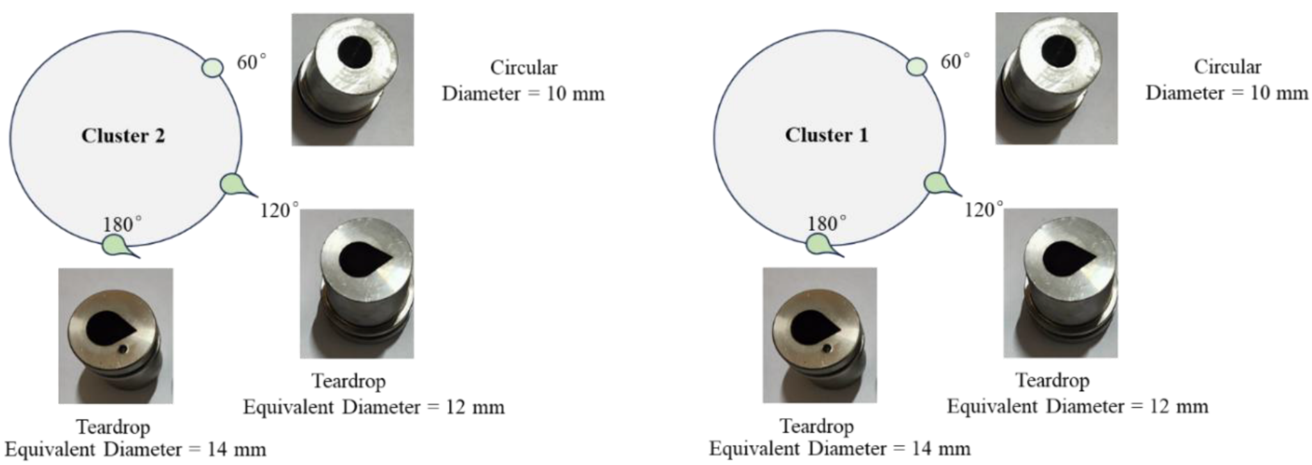

Figure 3—Perforations characteristics of different clusters

Experimental Scheme

In the Sichuan Basin shale gas reservoir, a single fracturing stage typically employs 72 perforations. Based on literature review, it is assumed that 36 perforations remain open before introducing temporary plugging materials. Field operations generally pump these materials at 3–6 m^3/min. Consequently, the average injection rate per open perforation during pumping is 0.083 m^3/min. This experiment uses 6 open perforations. To ensure the average injection rate per perforation aligns with field conditions, the experimental pump rate is set to 0.5 m^3/min. The study investigates the effect of temporary plugging materials dosage on perforation plugging by testing quantities equivalent to 1x, 1.5x, and 2x the number of open perforations. Additionally, for a given ball dosage, the impact of addition frequency is examined by introducing the balls in 1, 2, or 3 batches. Following ball addition (without stopping the pump), temporary plugging particulate (concentration: 4 kg/m^3; total quantity: 1 kg) is introduced. This evaluates the combined effect of ball and particulate on perforation plugging. The experimental design parameters are summarized in Table 1.

Table 1—Experimental Scheme

No	Open Perforations	Materials	Injection rate (m^3/min)	Viscosity (mPa.s)	Ball-to-Perforation Ratio	Dosage	Addition frequency
1	6	Balls	400	3	0.5	3	1
2	6	Balls	400	3	1	6	1
3	6	Balls	400	3	1	6	2(3/3)
4	6	Balls	400	3	1.5	9	1
5	6	Balls	400	3	1.5	9	2 (6/3)
6	6	Balls	400	3	1.5	9	3 (3/3/3)
7	6	Balls	400	3	2	12	1
8	6	Balls	400	3	2	12	2 (6/6)
9	6	Balls	400	3	2	12	3(4/4/4)
10	6	Balls & particulate	400	3	1	6	1
11	6	Balls & particulate	400	3	1.5	9	3(3/3/3)
12	6	Balls & particulate	400	3	2	12	3(4/4/4)
13	6	Knots	400	3	1	6	1
14	6	Knots	400	3	1.5	9	3(3/3/3)
15	6	Knots & particulate	400	3	1	6	1
16	6	Knots & particulate	400	3	1.5	9	3

Note: The Ball-to-Perforation Ratio is defined as the quotient of the number of temporary plugging balls or knots deployed divided by the count of open perforations

Experimental Procedure

(1) Close Valves 1 and 2. Load temporary plugging materials into the injection system according to preset parameters. (2) Initiate pumping at the designated rate. Monitor real-time injection rates through all perforations using the control/recording system. After flow stabilization, open Valve 2 to introduce materials into the horizontal wellbore. (3) For multi-stage material injections, maintain constant pumping rate during subsequent material introductions after the first stage. (4) For combined ball/particulate tests: After injecting balls, continuously pump particulate slurry without rate adjustment. (5) Post-injection, monitor until perforation flows stabilize. Shut down pumping system. Open Valve 1 to flush the horizontal wellbore.

Experimental Results

Plugging Process Characterization

Perforation injection rates were monitored in real-time to characterize plugging behavior: complete plugging was defined by 0 L/min flow, partial plugging by reduced flow (>0 L/min), and unplugged status by stable or increased flow.

Fig. 4 displays injection rate profiles for Test 11: (a) Cluster 2 and (b) Cluster 1 perforation responses. Initial deployment of three balls reduced flow at the Cluster 1-120° teardrop perforation from 72 to 15 L/min and the Cluster 2-180° teardrop perforation from 77 to 10 L/min, indicating partial sealing while other perforations exhibited flow increases. Maintaining a constant injection rate of 0.5 m³/min, a subsequent injection of three balls decreased flow at Cluster 2-120° from 94 to 30 L/min and Cluster 2-180° from 106 to 33 L/min, confirming persistent incomplete sealing at teardrop geometries. The third ball injection achieved complete flow termination at Cluster 1-60° circular perforation, with flow declining from 166 to 0 L/min, demonstrating geometry-dependent sealing efficacy. Subsequent particulate injection further reduced flows: Cluster 1-120° declined from 38 to 11 L/min (71% reduction), Cluster 1-180° from 38 to 8 L/min (79% reduction), Cluster 2-120° from 40 to 26 L/min (35% reduction), and Cluster 2-180° from 42 to 16 L/min (62% reduction). Despite these reductions, sustained flow through all teardrop perforations confirmed geometry-limited sealing efficiency. Concurrent flow augmentation at Cluster 2-60° evidenced preferential pathway development.

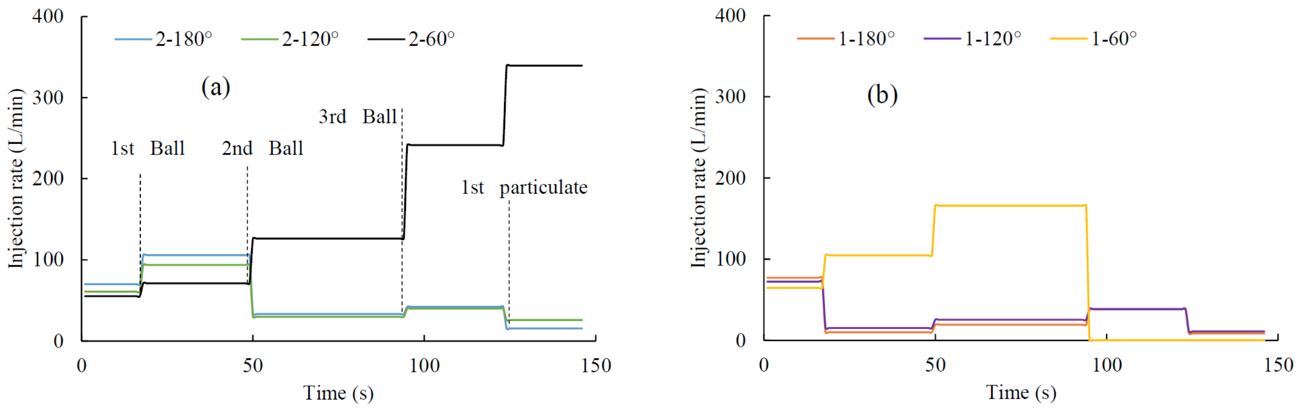

Figure 4—Injection Rate Variation Curve for Test Case 11

Fig. 5 displays injection rate profiles for Test 16. Initial deployment of three knots achieved complete flow termination at both Cluster 1-120° and 180° teardrop perforations, with flows declining from 72 to 0 L/min and 77 to 0 L/min respectively. Maintaining a constant injection rate of 0.5 m³/min, a subsequent fiber injection eliminated flow at the Cluster 1-60° circular perforation, reducing it from 90 to 0 L/min. The tertiary knots deployment produced complete flow cessation at Cluster 2-180°, where flow decreased from 108 to 0 L/min. Subsequent particulate injection demonstrated no measurable impact on Cluster 2-60° and 120° perforations, as maintained pre-injection flow rates confirmed ineffective sealing at these locations.

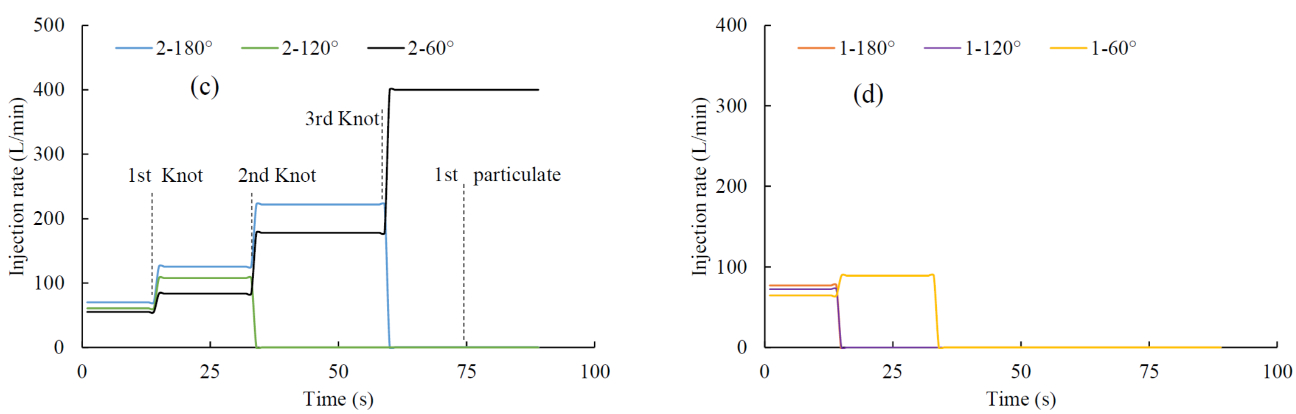

Figure 5—Injection Rate Variation Curve for Test Case 16

Perforation plugging sequencing, determined by flow rate reduction order, indicates that temporary plugging materials first seal toe-cluster perforations on the lower gravitational side, followed by either heel-cluster perforations on the lower side or toe-cluster perforations on the upper side, with heel-cluster upper-side perforations plugged last. Flow reduction magnitude analysis reveals: balls achieve complete plugging in circular perforations but prove ineffective at sealing tear-drop perforations. knots effectively seal both circular and tear-drop perforations. particulate alone demonstrate limited perforation plugging capability, however, when complemented with balls, they significantly enhance sealing efficiency in tear-drop perforations.

Spatial Plugging Distribution Characteristics

Fig. 6 quantifies plugging efficiency across perforations, revealing that Cluster 1 perforations at 120° and 180° phasing exhibit 100% plugging efficiency—indicating optimal susceptibility to temporary plugging materials. Conversely, Cluster 2 perforations at 60° phasing demonstrate the lowest probability (18.8%) across all tests, signifying maximal plugging resistance. Comparative analysis shows intermediate efficiency: 43.8% for Cluster 1 at 60° phasing, 56.3% for Cluster 2 at 120° phasing, and 68.8% for Cluster 2 at 180° phasing.

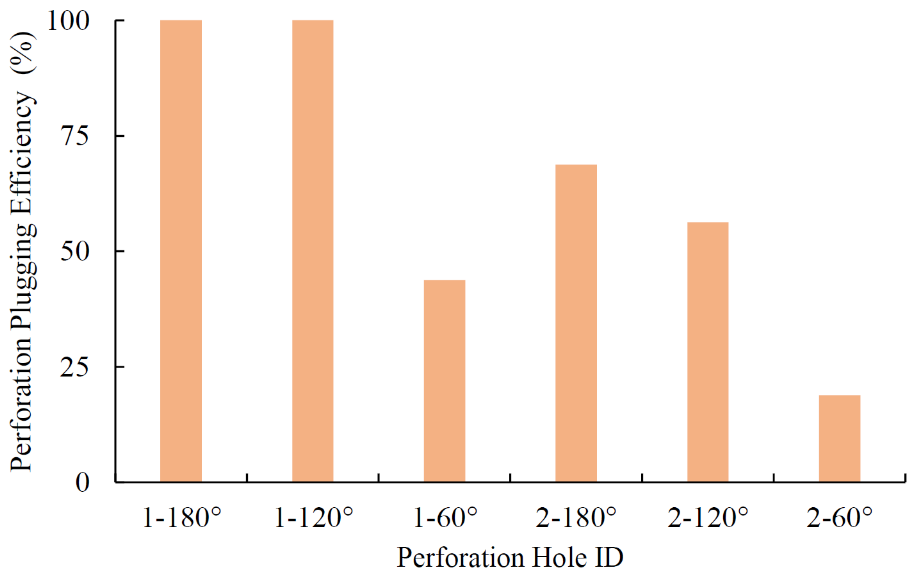

Figure 6—Perforation Plugging Efficiency

Fig. 7 presents experimental results of ball and particulate performance. Subfigure 7(a) shows that at a Ball-to-Perforation Ratio of 0.5 with single-stage injection, only Cluster 1-120° and 180° perforations were plugged while four remained unsealed. Increasing the ratio to 1.0 (7b) yielded identical sealing outcomes, indicating no incremental benefit. However, implementing two-stage injection at ratio 1.0 (7c) achieved sealing at both Cluster 1 and 2 (120°/180° positions), demonstrating that staged deployment enhances plugging efficiency without increasing material volume. At ratio 1.5 with single injection (7e), sealing expanded to all Cluster 1 perforations plus Cluster 2-180°, confirming that higher ratios improve coverage. Staged injections at this ratio (7f-g) both sealed five perforations, revealing diminishing returns beyond two deployment stages. Further increasing to ratio 2.0 (7i-k) produced identical five-perforation sealing as ratio 1.5, establishing 1.5 as the saturation threshold for ball-based diversion. Hybrid applications (7d; 7h; 7l) maintained equivalent sealing counts to ball-only treatments, yet crucially - as corroborated by Fig.3 - particulate enhanced sealing integrity at non-circular perforations despite lacking standalone plugging capability, highlighting their complementary role in complex geometries.

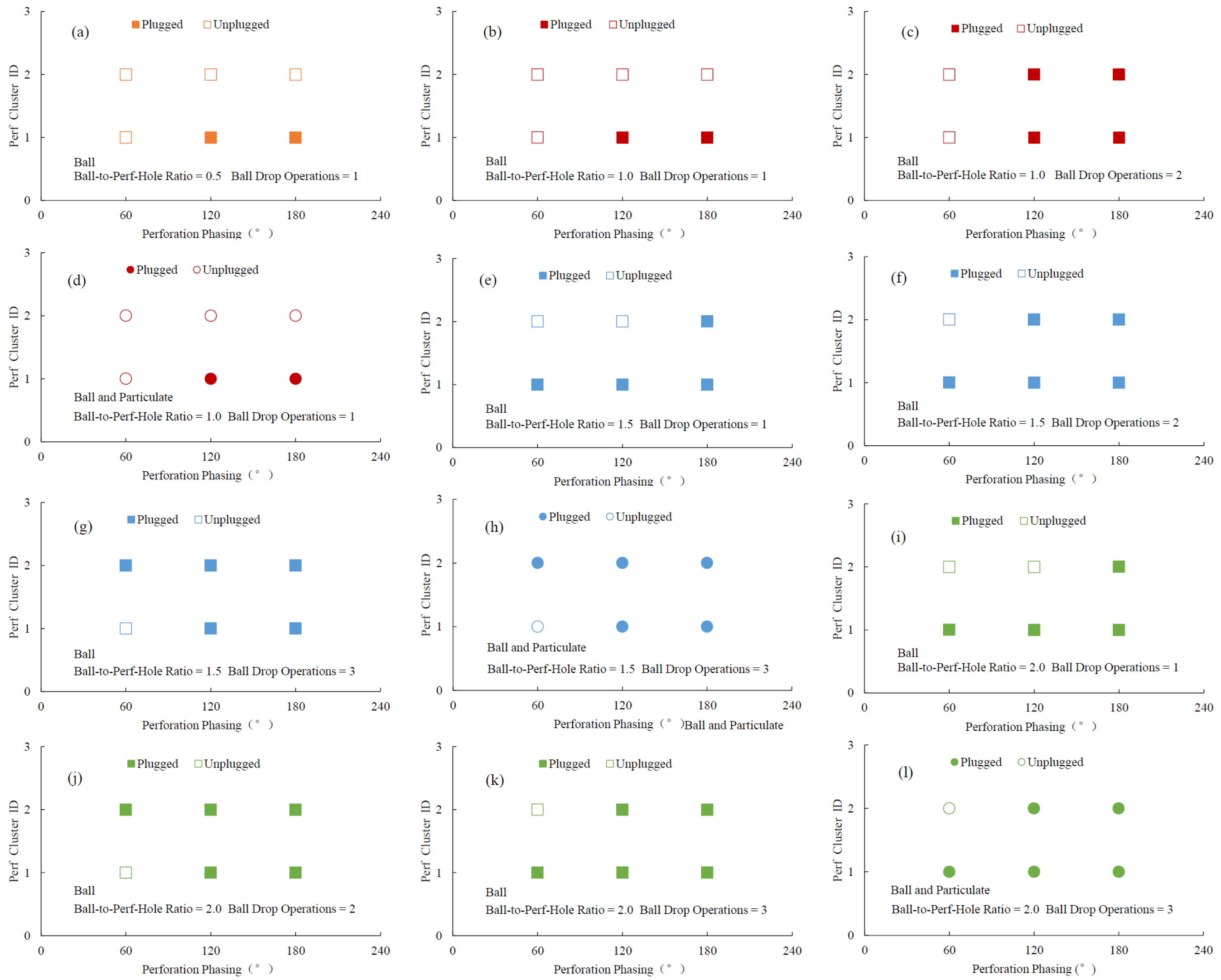

Figure 7—Experimental Results of Temporary plugging Ball

As depicted in Fig 8(a) and 8(c), a single injection of knots at Ball-to-Perforation Ratio 1.0 achieved sealing exclusively at Cluster 1-120° and 180° phased perforations. Subsequent particulate introduction maintained identical sealing counts without expanding coverage. Conversely, Fig.8(b) and 8(d) demonstrate that three-stage fiber deployment at ratio 1.5 sealed all perforations except cluster 2-60°, with particulate exerting no influence on the established sealing configuration - corroborating earlier findings that supplementary particulate primarily enhance sealing integrity rather than extend spatial coverage under optimized knots deployment protocols

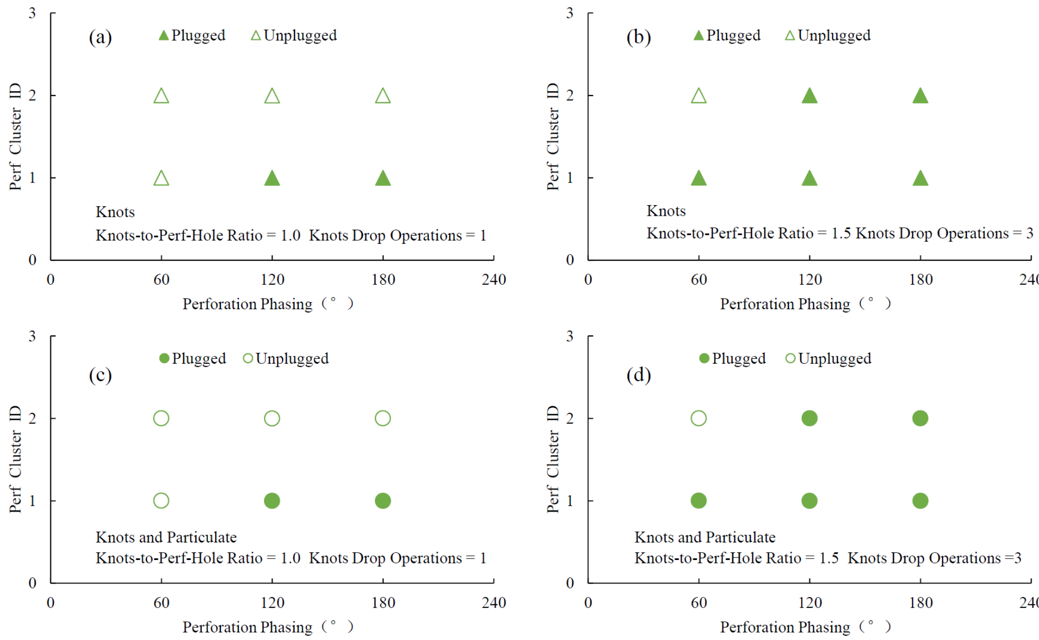

Figure 8—Experimental Results of Temporary plugging Knots

Field Application

Field implementation was conducted in Sichuan Basin's Well W204H49-3. This horizontal well features a 2750m true vertical depth with a 2000-meter lateral section segmented into 21 fracturing stages. Stage lengths ranged from 71 to 109 meters with cluster spacing between 6 and 11 meters. Each stage comprised six to seven clusters containing six perforations per cluster, resulting in 36 to 42 perforations per stage. Treatment parameters included fluid intensity of 27.8 m^3/m, proppant intensity of 2.8 t/m, and pump rates sustained at 16 to 18 m^3/min. Fig. 9 presents the Stage 14 fracturing treatment plot. Knots were deployed for inter-cluster diversion across 42 perforations. Pre-diversion step-rate flowback testing identified 20 open perforations, corresponding to a 47.6% opening ratio. The optimized knots dosage amounted to 1.5 times the open perforation count, totaling 30 units. These were injected in two discrete batches of 15 units each at 6 m^3/min to ensure perforation sealing. Post-stimulation step-rate testing confirmed 33 open perforations with a 78.6% opening ratio. This 31-percentage-point increase demonstrates significant inter-cluster diversion effectiveness.

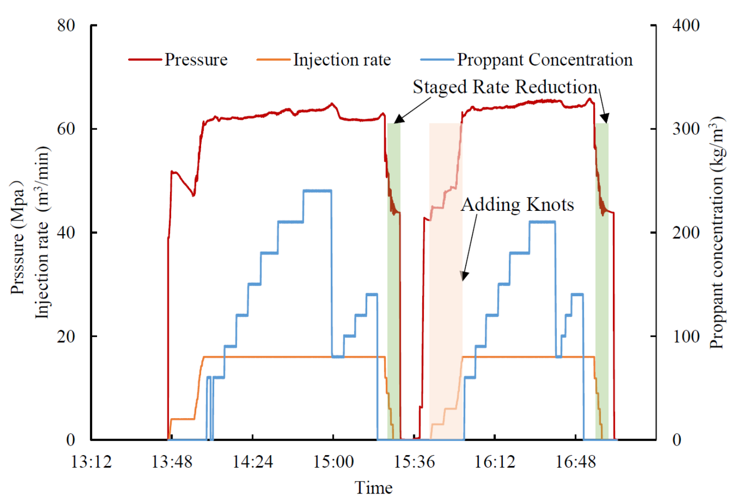

Figure 9—Stage 14 Fracturing Treatment Plot: Well W204H49-3

Conclusions

This study investigates the plugging characteristics of multiple temporary plugging material types under varying dosages and deployment methods in multi-cluster perforation scenarios, with field validation conducted in fracturing operations. Key findings are:

1. Temporary plugging materials initiate sealing preferentially at toe-side, gravity low-side perforations (120°/180° phasing), followed by either heel-side low-side perforations or toe-side high-side perforations (60° phasing), with heel-side high-side perforations sealed last.
2. Under constant material dosage, staged injection enhances plugging efficiency. At fixed injection stages, increasing the Ball-to-Perforation Ratio improves plugging efficiency until reaching saturation at approximately 1.5, beyond which no significant gains occur.
3. Temporary plugging balls: Achieve complete sealing at circular perforations but only partial sealing at teardrop-shaped perforations. Temporary plugging knots: Deliver complete sealing independent of geometry. Temporary plugging particulate: Synergistically reduce flow at ball-sealed teardrop perforations but cannot achieve full plugging.

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